Where vision-language models fail at 3D space: a VSI-Bench evaluation

Qwen3-VL-8B-Instruct · Gemma-4-12B-it · Molmo2-8B | VSI-Bench, 1316 questions over 288 scenes | Multimodal Computer Vision — HW

SETUP AND PROTOCOL

Item	Setting
Benchmark	VSI-Bench (CVPR 2025): 5,130 QA over 288 egocentric videos from ScanNet / ScanNet++ / ARKitScenes val. Videos ship with the public dataset — no ScanNet registration required.
Tasks (full sets)	`route_planning` (Cat. 1, ACC, n=194) · `object_abs_distance` (Cat. 2, MRA, n=834) · `room_size_estimation` (Cat. 2, MRA, n=288, one per scene)
Models (Apache-2.0)	Qwen3-VL-8B-Instruct · Gemma-4-12B-it · Molmo2-8B — local inference, bf16, single H100 80 GB
Visual input	32 frames at `round(linspace(0, N−1, 32))`, long side ≤448 px, fed as a multi-image stack to every model (no native video path, so the input is identical across models)
Decoding / prompts	greedy (`do_sample=False`), `max_new_tokens=48`, Gemma-4 thinking disabled; prompts verbatim from the official `vsibench_doc_to_text()`
Metrics / control	official ACC & MRA copied verbatim; lenient unit-aware scorer as a secondary check; 95% CI by scene-clustered bootstrap; blind arm (no frames) as a control

RESULTS

Model	Route planning ACC · n=194	Absolute distance MRA · n=834	Room size MRA · n=288
Qwen3-VL-8B-Instruct	35.0 [28, 42]	47.9 [45, 51]	57.1 [53, 61]
Gemma-4-12B-it	30.4 [24, 37]	0.3 [0, 1] lenient 22.8 (+22.4)	0.0 [0, 0]
Molmo2-8B	29.4 [23, 36]	0.0 [0, 0] lenient 16.2 (+16.2)	11.5 [8, 14] lenient 18.6 (+7.1)

ACC = exact match on the option letter; MRA = Mean Relative Accuracy (official implementations, copied verbatim). Bracketed values are 95% confidence intervals from a bootstrap that resamples scenes. Grey "lenient" figures come from a secondary unit-aware parser and show how much of the score is answer formatting rather than spatial reasoning.

Numeric failure signatures

Model	Task	r(pred, GT)	$\sigma_{pred}/\sigma_{GT}$	Parse failures	Median rel. error
Gemma-4-12B-it	Absolute distance	0.2341	0.6624	96.8%	0.54
Molmo2-8B	Absolute distance	0.3332	2.9955	99.9%	1.0
Molmo2-8B	Room size	0.7425	3.2377	38.2%	0.7085
Qwen3-VL-8B-Instruct	Absolute distance	0.7143	0.6487	0.0%	0.2753
Qwen3-VL-8B-Instruct	Room size	0.8136	0.5758	0.0%	0.2066

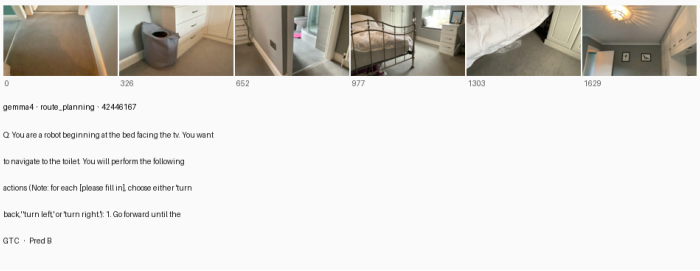
A model that measures a room should have r(pred, GT) well above zero and spread near the ground truth; r ≈ 0 with $\sigma_{pred}/\sigma_{GT} \ll 1$ is prior recall.

Control arm: same questions, no frames

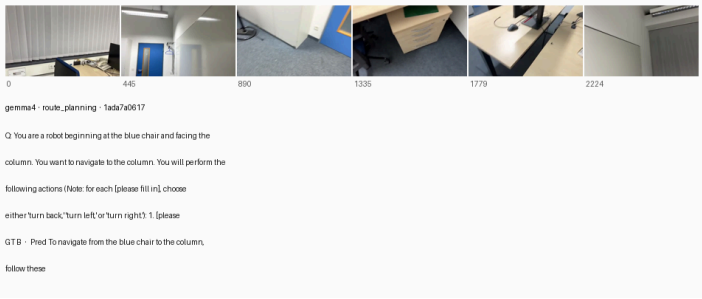
Model	Absolute distance full · blind · Δ	Room size full · blind · Δ
Qwen3-VL-8B-Instruct	47.9 · 30.6 · **+17.4**	57.1 · 32.9 · **+24.2**
Gemma-4-12B-it	0.3 · 0.0 · **+0.3**	0.0 · 0.0 · **+0.0**
Molmo2-8B	0.0 · 0.1 · **-0.1**	11.5 · 32.0 · **-20.6**

REPRESENTATIVE FAILURE CASES

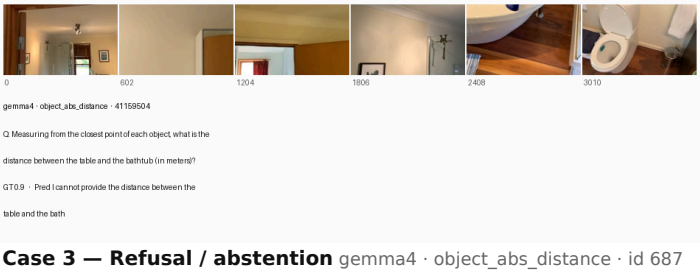
Each panel shows 6 of the 32 frames the model actually saw (labelled with their source frame index), the question, the ground truth, and the raw prediction.



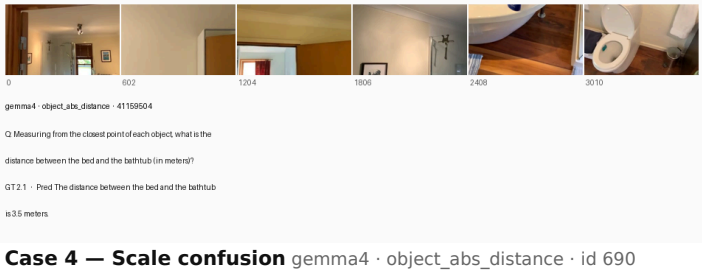
Case 1 — Egocentric direction gemma4 · route_planning · id 4959



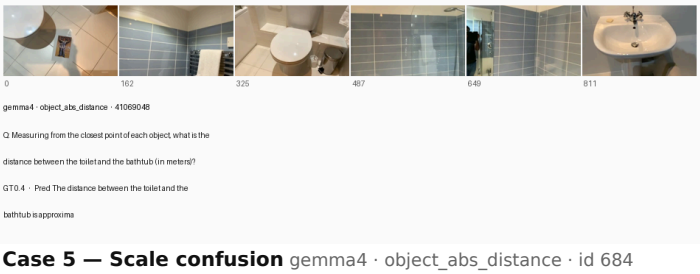
Case 2 — Answer format gemma4 · route_planning · id 4963



Case 3 — Refusal / abstention gemma4 · object_abs_distance · id 687



Case 4 — Scale confusion gemma4 · object_abs_distance · id 690



Case 5 — Scale confusion gemma4 · object_abs_distance · id 684

FAILURE TAXONOMY

Class	n	Mechanism
Scale confusion	1824	Large metric error: the model commits to a plausible-looking magnitude the scene does not support. Prior-pinned answers are the subset of these that land near the task's own median.
Egocentric direction	378	Wrong turn sequence in route planning: a left/right mirror, a failure to accumulate heading across frames, or a letter prior (B and C over-predicted, A and D under-used).
Refusal / abstention	355	The model declines to answer. Not a scale error and not a formatting artifact: no estimate is produced at all.
Answer format	232	Right in substance but invisible to the official first-token parser (prose, unit change, turn/channel markers). Quantified by the strict-vs-lenient gap, not by the failure gallery.

2789 failing questions across all models and non-blind tasks (near misses inside 30% relative error are excluded: MRA already gives partial credit).

DISCUSSION

Qwen3-VL-8B is the only model of the three that measures rather than recites, reaching MRA 0.479 on absolute distance and 0.571 on room size against 0.350 route-planning accuracy. Removing the video costs it 17.3 and 24.2 MRA points with non-overlapping confidence intervals, yet roughly a third of its distance score survives with no image at all, so part of the score is prior recall in the benchmark as much as in the model. Gemma-4-12B returns no room-size estimate on any of the 288 questions, and its absolute-distance answers need the lenient parser to reach MRA 0.228, so its 0.3 strict figure measures answer format rather than spatial ability. Molmo2-8B's room-size answers correlate with ground truth ($r = 0.74$) but carry three times its spread, and its blind arm — a constant 20 m² — lands within one point of the best constant any model could score against this ground truth, which is a property of MRA's threshold structure as much as of the model. Route planning is where all three converge (29–35%) and the confusion matrix shows a shared mechanism: B and C are over-predicted and A and D under-used, so much of the error is a letter prior rather than a failed turn. The largest remaining gain is in the input, not the model: 32 frames at 448 px cannot carry the metric scale of a room, and asking the model to verify a proposed depth or floor-plan estimate would test spatial reasoning instead of recall. The limitation is structural — no frame budget repairs a benchmark whose numeric answers can be scored above chance by a constant.

REFERENCES

1. J. Yang, S. Yang, A. Gupta, R. Han, L. Fei-Fei, S. Xie. *Thinking in Space: How Multimodal Large Language Models See, Remember and Recall Spaces.* CVPR 2025. arXiv:2412.14171. Dataset and harness: huggingface.co/datasets/nyu-visionx/VSI-Bench; github.com/vision-x-nyu/thinking-in-space.
2. Qwen Team. *Qwen3-VL* model card. huggingface.co/Qwen/Qwen3-VL-8B-Instruct.
3. Google DeepMind. *Gemma 4* model card. huggingface.co/google/gemma-4-12B-it.
4. Allen Institute for AI. *Molmo 2* model card. huggingface.co/allenai/Molmo2-8B.
5. Z. Fan et al. *VLM-3R: Vision-Language Models Augmented with Instruction-Aligned 3D Reconstruction.* CVPR 2026. arXiv:2505.20279. Scenes from ScanNet (CVPR 2017) and ScanNet++ (ICCV 2023).

Protocol hash `517a77ed69be` · frames 32 · greedy · all results from this harness.